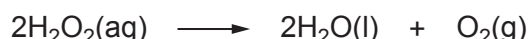


7. A catalyst is a substance that causes significant changes to the rate of a chemical reaction. Catalysts interact with the reactants in a chemical reaction and eventually release the product allowing the catalyst to be recovered.

Enzymes are biological molecules which catalyse reactions in living organisms. Enzymes tend to have an optimum temperature of 37 °C.

Hydrogen peroxide is a colourless liquid with the chemical formula H_2O_2 . It has many uses including as a bleach and an antiseptic. It is also found in all animals and plants. The decomposition of hydrogen peroxide happens very slowly on its own.



The reaction can be catalysed by several substances including iron(III) oxide, Fe_2O_3 , manganese dioxide, MnO_2 , and the enzyme catalase.

Four catalysts were tested to determine which was the most efficient. An equal mass of each catalyst was added to equal volumes of hydrogen peroxide solution of equal concentration. The volume of oxygen produced in 1 minute was recorded. The experiment was repeated at different temperatures.

Temperature (°C)	Volume of oxygen produced in 1 minute (cm ³)			
	Catalyst W	Catalyst X	Catalyst Y	Catalyst Z
10	0.8	1.2	0.4	0.9
20	1.4	1.4	0.9	1.5
30	1.9	2.1	1.3	2.1
40	2.5	2.6	1.6	2.5
50	3.4	3.8	1.2	3.5
60	4.6	4.8	0.7	4.7
70	5.2	5.4	0.3	5.3



(a) Give the **number** of the statement which is **not** correct.

[1]

- 1 Catalyst **W** is more efficient than **Y** but less efficient than **X** and **Z**
 - 2 Catalyst **X** is the most efficient of the four
 - 3 Catalyst **Y** is less efficient than all of the others
 - 4 Catalyst **Z** is more efficient than **Y** but less efficient than **W** and **X**
-

(b) State which catalyst, **W**, **X**, **Y** or **Z**, could be catalase. Explain your answer.

[3]

Catalyst

.....

.....

.....

(c) Three students made statements about the properties of catalysts.

- Katrina said that a catalyst does not get used up in the reaction but makes it happen faster.
- Jo said that the reactants are used up in less time because a catalyst makes a different product.
- Rhys said that adding a catalyst allows the reaction to happen with a lower activation energy, but the same products are made.

Which of the students' statements are correct?

[1]

- A** Jo only
- B** Rhys and Jo, but not Katrina
- C** Katrina and Rhys, but not Jo
- D** All three students
-

